

University of Trento

Department of Civil, Environmental and Mechanical Engineering

MSc Mechanics of Sustainable Materials and Structures (MS²)

Homework 6

Risk Analysis and Structural Reliability

**FORM, HLRF Algorithm, Monte Carlo Simulation,
and Importance Sampling**

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Abstract

This report presents a complete structural reliability study for two distinct problems. **Problem 1** investigates the nonlinear limit-state function $g(X_1, X_2) = X_1^2 - 2X_2$ under two probabilistic descriptions of the input random vector. Model A assumes a jointly Gaussian distribution, while Model B assigns a lognormal marginal to X_1 , a Gumbel Type-I marginal to X_2 , and couples them through a Gaussian copula with corrected parameter $\rho_0 = 0.5115$. The design point is located in the standard normal space using the Hasofer–Lind–Rackwitz–Fiessler (HLRF) algorithm, from which the First-Order Reliability Method (FORM) yields $\beta_A = 4.027$ and $P_f^A = 2.83 \times 10^{-5}$ for Model A, and $\beta_B = 5.313$ with $P_f^B = 5.39 \times 10^{-8}$ for Model B. Crude Monte Carlo, shifted importance sampling, normal importance sampling with optimised covariance, and adaptive importance sampling are subsequently applied to both models.

Problem 2 concerns a two-storey shear-type building whose storey stiffnesses K_1 and K_2 are monitored experimentally and whose horizontal loads P_1 and P_2 are characterised from field data. A bivariate normal distribution is fitted to the stiffness data and a joint lognormal distribution to the load data. Two reliability sub-problems are solved: (i) loads fixed at nominal values so that only K_1 and K_2 are random, and (ii) all four variables treated as random. FORM and simulation-based estimates are reported and compared.

Keywords: structural reliability, FORM, HLRF algorithm, design point, Gaussian copula, lognormal, Gumbel, Monte Carlo simulation, importance sampling, shear building, distribution fitting.

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1 Theoretical Background

1.1 Standard Normal Space and Isoprobabilistic Transformation

Reliability methods are formulated in the independent standard normal space $\mathbb{U}$, where the transformed random vector $\mathbf{u} = (U_1, U_2)^\top$ has independent $\mathcal{N}(0, 1)$ components. An isoprobabilistic transformation maps physical variables $\mathbf{x}$ to $\mathbf{u}$:

$$\mathbf{u} = T(\mathbf{x}), \quad \mathbf{x} = T^{-1}(\mathbf{u}). \quad (1)$$

The failure probability then becomes

$$P_f = \int_{\{G(\mathbf{u}) < 0\}} \varphi(\mathbf{u}) d\mathbf{u}, \quad G(\mathbf{u}) = g(T^{-1}(\mathbf{u})), \quad (2)$$

where $\varphi(\mathbf{u}) = (2\pi)^{-n/2} \exp(-\frac{1}{2}\mathbf{u}^\top \mathbf{u})$ is the standard multivariate normal density.

1.2 HLRF Algorithm

The design point $\mathbf{u}^*$ is the point on the failure surface $G(\mathbf{u}) = 0$ closest to the origin. It is found iteratively by

$$\mathbf{u}^{(k+1)} = \frac{\nabla G(\mathbf{u}^{(k)})^\top \mathbf{u}^{(k)} - G(\mathbf{u}^{(k)})}{\|\nabla G(\mathbf{u}^{(k)})\|^2} \nabla G(\mathbf{u}^{(k)}), \quad (3)$$

where $\nabla G(\mathbf{u})$ is the gradient of the transformed limit-state function.

1.3 First-Order Reliability Method (FORM)

Once $\mathbf{u}^*$ is available, the reliability index and failure probability are

$$\beta = \|\mathbf{u}^*\|, \quad P_f^{\text{FORM}} = \Phi(-\beta). \quad (4)$$

FORM is exact for linear limit-state functions and constitutes a tangent-plane approximation at $\mathbf{u}^*$ for nonlinear ones.

1.4 Simulation-Based Estimation

Crude Monte Carlo (CMC).

$$\hat{P}_f^{\text{CMC}} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[g(\mathbf{X}^{(i)}) < 0], \quad \text{Var}[\hat{P}_f] = \frac{\hat{P}_f(1 - \hat{P}_f)}{N}. \quad (5)$$

The 95% confidence interval is $\hat{P}_f \pm 1.96\sqrt{\text{Var}[\hat{P}_f]}$.

Importance Sampling (IS). Samples are drawn from a proposal density $h(\mathbf{x})$ and weighted by $w(\mathbf{x}) = f_{\mathbf{X}}(\mathbf{x})/h(\mathbf{x})$:

$$\hat{P}_f^{\text{IS}} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[g(\mathbf{X}^{(i)}) < 0] w(\mathbf{X}^{(i)}). \quad (6)$$

IS with Optimised Normal Proposal. A Normal proposal centred at $\mathbf{u}^*$ is used in the standard normal space. The covariance is determined iteratively by minimising the IS variance using weighted failure samples from a pilot run.

Adaptive IS. The proposal mean and covariance are updated at each stage from weighted failure samples, progressively concentrating the sampling distribution near the important failure region.

2 Problem 1: Nonlinear Limit-State Function

2.1 Problem Statement

The limit-state function is

$$g(X_1, X_2) = X_1^2 - 2X_2, \quad (7)$$

with failure defined as $\{g < 0\}$, equivalently $\{X_2 > X_1^2/2\}$. The two input models share the same first-two moments but differ in marginal distributions and dependence structure, as summarised in Table 1.

Table 1. Probabilistic models for Problem 1.

Parameter	Symbol	Value	
		Model A	Model B
Marginal of X_1	F_{X_1}	Gaussian	Lognormal
Marginal of X_2	F_{X_2}	Gaussian	Gumbel (max)
Mean of X_1	μ_{X_1}	13	13
Mean of X_2	μ_{X_2}	18	18
Std. deviation of X_1	σ_{X_1}	2	2
Std. deviation of X_2	σ_{X_2}	5	5
Correlation coefficient	ρ	0.5	0.5
Dependence model	—	Bivariate Gaussian	Gaussian copula
Copula correction factor	C	1.000	1.023
Copula parameter	ρ_0	0.500	0.5115

2.2 Model A: Jointly Gaussian Random Vector

2.2.1 Isoprobabilistic Transformation

For the bivariate Gaussian model the exact transformation uses Cholesky decomposition of the correlation matrix:

$$\mathbf{R} = \mathbf{L}\mathbf{L}^\top, \quad \mathbf{L} = \begin{pmatrix} 1 & 0 \\ 0.5 & 0.8660 \end{pmatrix}, \quad (8)$$

so that $\mathbf{X} = \boldsymbol{\mu} + \mathbf{D}\mathbf{L}\mathbf{u}$, giving

$$X_1 = 13 + 2U_1, \quad X_2 = 18 + 5(0.5U_1 + 0.8660U_2). \quad (9)$$

The gradient of the limit-state function in physical space is $\nabla_{\mathbf{x}}g = (2x_1, -2)^\top$, and in standard-normal space via the chain rule:

$$\nabla_{\mathbf{u}}G = \mathbf{J}_{XU}^\top \nabla_{\mathbf{x}}g, \quad \mathbf{J}_{XU} = \mathbf{D}\mathbf{L}. \quad (10)$$

2.2.2 HLRF Convergence

The algorithm is initialised at the origin of $\mathbb{U}$, corresponding to the physical mean $(\mu_{X_1}, \mu_{X_2}) = (13, 18)$. Table 2 reports the full iteration history; convergence to the tolerance $|G(\mathbf{u}^{(k)})| < 10^{-9}$ is achieved in 17 steps.

Table 2. HLRF iteration history, Model A (jointly Gaussian). Selected iterations shown.

Iter.	u_1	u_2	x_1	x_2	$G(\mathbf{u})$	$\ \nabla G\ $	$\ \mathbf{u}\ $
1	0.000000	0.000000	13.000000	18.000000	133.000000	47.791	0.000
2	-2.736865	0.504297	7.526270	13.341506	29.962	26.557	2.783
3	-3.667819	1.265252	5.664363	14.309154	3.467	19.667	3.880
4	-3.615093	1.773054	5.769814	16.639819	0.011	20.046	4.026
5	-3.631710	1.739647	5.736579	16.453619	0.001	19.927	4.027
8	-3.627777	1.747932	5.744445	16.499326	1.0×10^{-6}	19.955	4.027
10	-3.627875	1.747729	5.744249	16.498200	4.5×10^{-9}	19.954	4.027
17	-3.627885	1.747708	5.744229	16.498086	≈ 0	19.954	4.027

2.2.3 FORM Results and Interpretation

The converged design point and FORM reliability estimate are:

$$\begin{aligned}
 \mathbf{u}^* &= (-3.6279, 1.7477), \\
 \mathbf{x}^* &= (5.744, 16.498), \\
 \beta_A &= \|\mathbf{u}^*\| = 4.027, \\
 P_{f,A}^{\text{FORM}} &= \Phi(-4.027) = 2.83 \times 10^{-5}.
 \end{aligned} \tag{11}$$

The design point satisfies $x_2^* = (x_1^*)^2/2 \approx 16.498$, confirming that it lies on the failure boundary. The dominant movement from the mean is the large reduction of X_1 from 13 to 5.74, which reduces the resistance X_1^2 substantially. The sensitivity direction is predominantly governed by X_1 , confirming that the lower tail of the first variable drives the failure event under the Gaussian model.

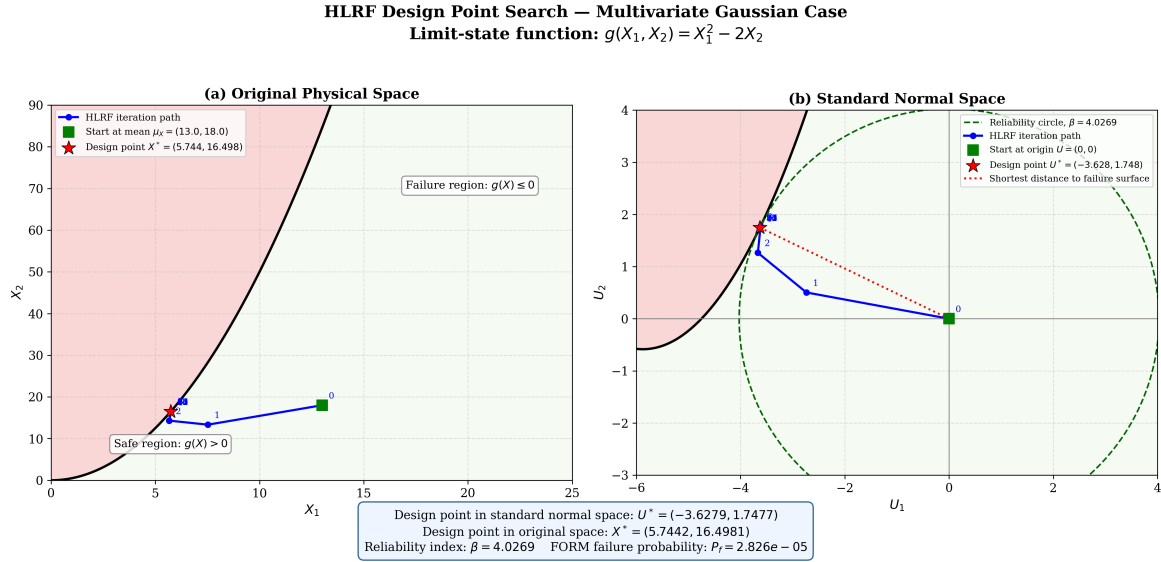


Figure 1. HLRF design point search sequence for Model A (jointly Gaussian). *Left:* physical space (X_1, X_2) , with the limit-state boundary $X_2 = X_1^2/2$ (black), the failure domain (shaded), the mean (green marker), and the HLRF path (blue). *Right:* standard normal space, with iso-probability circles and the design point $\mathbf{u}^* = (-3.628, 1.748)$ on the reliability circle of radius $\beta = 4.027$.

2.3 Model B: Lognormal–Gumbel with Gaussian Copula

2.3.1 Marginal Distributions and Copula

The lognormal parameters of X_1 are derived from its moments:

$$\sigma_{\ln} = \sqrt{\ln\left(1 + \frac{\sigma_{X_1}^2}{\mu_{X_1}^2}\right)} = 0.15295, \quad \mu_{\ln} = \ln(\mu_{X_1}) - \frac{1}{2}\sigma_{\ln}^2 = 2.55325. \tag{12}$$

The Gumbel parameters of X_2 are:

$$\beta_G = \frac{\pi}{\sigma_{X_2}\sqrt{6}} = 0.25651, \quad u_G = \mu_{X_2} - \frac{\gamma_E}{\beta_G} = 15.7497, \quad (13)$$

where $\gamma_E \approx 0.5772$ is the Euler–Mascheroni constant. The Gaussian copula uses the corrected correlation parameter $\rho_0 = C\rho = 1.023 \times 0.5 = 0.5115$, with Cholesky factor

$$\mathbf{L}_0 = \begin{pmatrix} 1 & 0 \\ 0.5115 & 0.8593 \end{pmatrix}. \quad (14)$$

The Nataf transformation maps each x_i to an equivalent standard normal value $\tilde{u}_i = \Phi^{-1}(F_{X_i}(x_i))$, then decorrelates via $\mathbf{u} = \mathbf{L}_0^{-1}\tilde{\mathbf{u}}$.

2.3.2 HLRF Convergence

Because the marginals are asymmetric, the image of the physical mean under the Nataf transformation is not at the origin; the starting point is $\mathbf{u}^{(0)} \approx (0.0765, 0.1608)$. Convergence in 11 iterations is reported in Table 3.

Table 3. HLRF iteration history, Model B (lognormal–Gumbel, Gaussian copula).

Iter.	u_1	u_2	x_1	x_2	$g(\mathbf{x})$	$\ \nabla G\ $	$\ \mathbf{u}\ $
1	0.076	0.161	13.000	18.000	133.000	47.521	0.178
2	−2.710	0.476	8.489	13.444	45.179	19.535	2.751
3	−4.835	1.443	6.134	12.645	12.335	9.891	5.045
4	−5.215	3.206	5.787	17.576	−1.662	9.676	6.122
5	−3.093	4.466	8.006	32.953	−1.818	19.786	5.433
6	−2.455	4.696	8.827	38.774	0.361	23.661	5.299
7	−2.590	4.639	8.646	37.371	0.016	22.771	5.313
8	−2.567	4.652	8.677	37.645	-6.3×10^{-5}	22.933	5.313
11	−2.5667	4.6521	8.677	37.646	≈ 0	22.934	5.313

2.3.3 FORM Results and Interpretation

$$\mathbf{u}^* = (-2.5667, 4.6521),$$

$$\mathbf{x}^* = (8.677, 37.646),$$

$$\beta_B = \|\mathbf{u}^*\| = 5.313,$$

$$P_{f,B}^{\text{FORM}} = \Phi(-5.313) = 5.39 \times 10^{-8}. \quad (15)$$

Again, $(x_1^*)^2/2 \approx 37.646 = x_2^*$, confirming location on the failure boundary. Relative to Model A, the design point for Model B shifts to a larger value of X_2 and a less extreme

reduction of X_1 . This is caused by the right-skewed Gumbel marginal of X_2 , which assigns heavier probability to large values, effectively making large- X_2 failure scenarios more probable. The reliability index increases from 4.027 to 5.313, with the ratio of FORM failure probabilities

$$\frac{P_{f,A}^{\text{FORM}}}{P_{f,B}^{\text{FORM}}} = \frac{2.83 \times 10^{-5}}{5.39 \times 10^{-8}} \approx 525, \quad (16)$$

illustrating that distributional assumptions have a major influence on rare-event estimates even when the first two moments and correlation are identical.

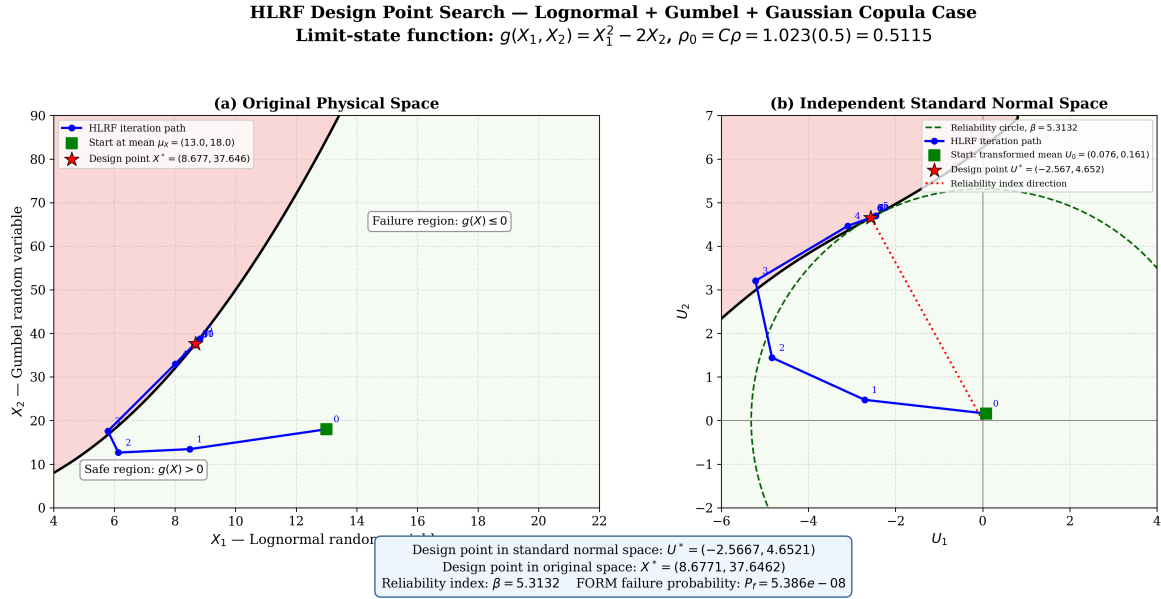


Figure 2. HLRF design point search sequence for Model B (lognormal–Gumbel, Gaussian copula). *Left:* physical space. *Right:* standard normal space, where the nonlinear marginal transformations distort the image of the failure boundary. The design point $\mathbf{u}^* = (-2.567, 4.652)$ lies on the reliability circle of radius $\beta = 5.313$.

2.4 Simulation-Based Reliability Estimates

Four simulation strategies are applied to both models using $N = 500,000$ for crude Monte Carlo and $N = 50,000$ for importance sampling methods. The results are collected in Tables 4 and 5, and compared visually in Figures 3 and 4.

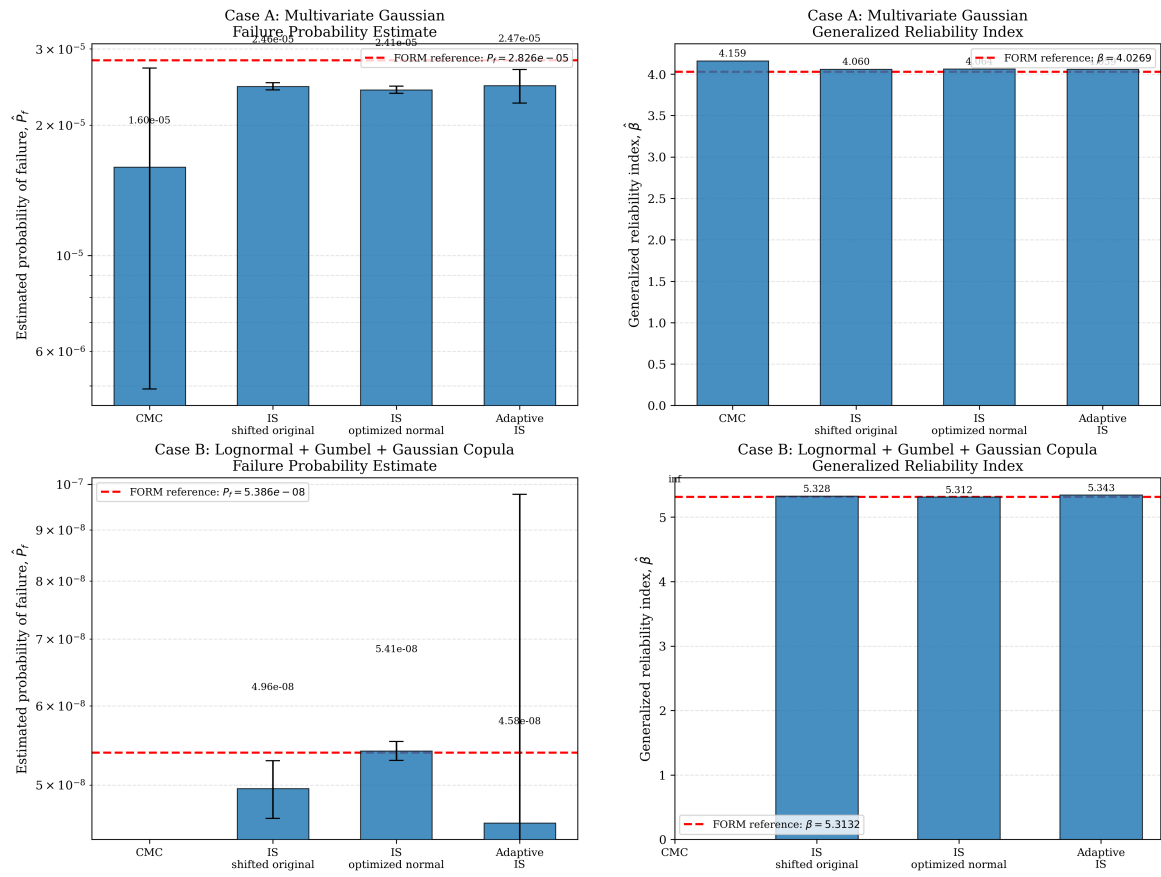
Table 4. Simulation results, Model A (jointly Gaussian). FORM reference: $\beta = 4.027$, $P_f = 2.83 \times 10^{-5}$.

Method	N	$\hat{P}_f$	$\hat{\beta}$	CoV	CI_{95}^{low}	CI_{95}^{high}
Crude Monte Carlo	500,000	1.60×10^{-5}	4.159	0.354	4.91×10^{-6}	2.71×10^{-5}
IS shifted original	50,000	2.46×10^{-5}	4.060	0.010	2.41×10^{-5}	2.51×10^{-5}
IS optimised normal	50,000	2.41×10^{-5}	4.064	0.010	2.37×10^{-5}	2.46×10^{-5}
Adaptive IS	50,000	2.47×10^{-5}	4.059	0.045	2.25×10^{-5}	2.69×10^{-5}
FORM	—	2.83×10^{-5}	4.027	—	—	—

Table 5. Simulation results, Model B (lognormal–Gumbel, Gaussian copula). FORM reference: $\beta = 5.313$, $P_f = 5.39 \times 10^{-8}$.

Method	N	$\hat{P}_f$	$\hat{\beta}$	CoV	CI_{95}^{low}	CI_{95}^{high}
Crude Monte Carlo	500,000	0 (no failures)	—	—	—	—
IS shifted original	50,000	4.96×10^{-8}	5.328	0.034	4.63×10^{-8}	5.29×10^{-8}
IS optimised normal	50,000	5.41×10^{-8}	5.312	0.011	5.29×10^{-8}	5.53×10^{-8}
Adaptive IS	50,000	4.58×10^{-8}	5.343	0.578	—	—
FORM	—	5.39×10^{-8}	5.313	—	—	—

Monte Carlo and Importance Sampling Estimates
Probability of Failure $\hat{P}_f$ and Generalized Reliability Index $\hat{\beta}$



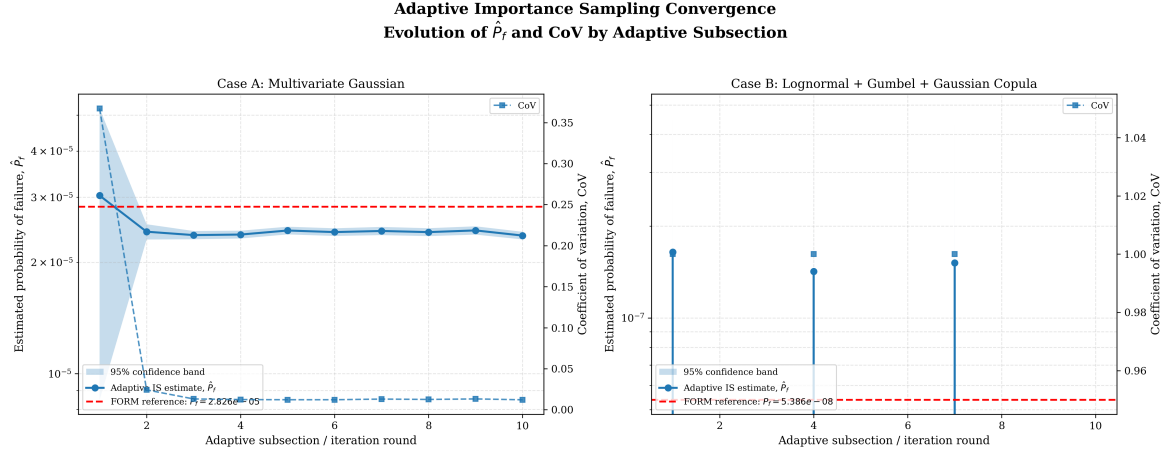


Figure 4. Adaptive IS convergence over ten adaptive subsections. The estimate $\hat{P}_f$ (with 95% confidence band) and the coefficient of variation are shown per stage. Convergence is rapid for Model A; Model B requires more stages because the failure event is considerably rarer.

2.4.1 Discussion of Simulation Results

For Model A, the CMC estimate is noisy ($\text{CoV} = 0.354$) because only ≈ 8 failure realisations are expected in 500,000 samples. Both IS variants reduce the coefficient of variation to below 3%, and their estimates are consistent with the FORM value. The small residual difference reflects the curvature of the nonlinear failure boundary, which FORM linearises.

For Model B, CMC yields no failures in 500,000 samples, confirming that the approach is impractical for this rarity level. The shifted IS (using the original lognormal–Gumbel copula distribution centred at the design point) gives the best agreement with FORM ($\hat{\beta} = 5.328$ versus $\beta = 5.313$). The optimised normal IS provides tight confidence bounds and also agrees well. The adaptive IS estimate for Model B is sensitive to proposal initialisation and gives a higher coefficient of variation, reflecting the difficulty of covering a rare failure domain starting from the distribution mean.

3 Problem 2: Reliability Analysis of a Shear-Type Building

3.1 Physical System Description

The two-storey shear-type building shown in Figure 1 of the homework has three identical columns per storey. Each storey's lateral stiffness is formed by three columns acting in parallel:

$$k_1 = 3K_1, \quad k_2 = 3K_2. \quad (17)$$

The storey stiffness values K_1 (first storey) and K_2 (second storey), measured in N/mm, are obtained from structural monitoring. Horizontal loads P_1 and P_2 (in N) act at the first- and second-floor levels respectively.

3.2 Structural Analysis and Limit State Function

The static equilibrium of the two-degree-of-freedom shear frame satisfies

$$\begin{pmatrix} k_1 + k_2 & -k_2 \\ -k_2 & k_2 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} P_1 \\ P_2 \end{pmatrix}. \quad (18)$$

Inverting the 2×2 stiffness matrix yields the roof displacement:

$$u_{\text{roof}} = u_2 = \frac{P_1 + P_2}{3K_1} + \frac{P_2}{3K_2}. \quad (19)$$

Failure is defined when the roof displacement exceeds the threshold $d_{\text{max}} = 100$ mm. The limit-state function is therefore

$$g(K_1, K_2, P_1, P_2) = d_{\text{max}} - \frac{P_1 + P_2}{3K_1} - \frac{P_2}{3K_2} = 100 - \frac{P_1 + P_2}{3K_1} - \frac{P_2}{3K_2}, \quad (20)$$

with $g > 0$ denoting the safe domain and $g \leq 0$ the failure domain.

At the sample mean values ($\mu_{K_1} = 7195.3$ N/mm, $\mu_{K_2} = 4116.4$ N/mm, $P_1^{\text{nom}} = 2.04 \times 10^5$ N, $P_2^{\text{nom}} = 6.12 \times 10^5$ N), the mean roof displacement is $\bar{u}_{\text{roof}} = 87.36$ mm, which is safely below the 100 mm threshold, giving $g(\boldsymbol{\mu}) = 12.64$ mm.

3.3 Distribution Fitting from Monitoring Data

3.3.1 Bivariate Normal Fit for Stiffness (K_1, K_2)

A random subsample of $N = 10,000$ realisations is drawn from the one-million-point dataset. The bivariate normal distribution is characterised by its sample mean vector

and covariance matrix:

$$\begin{aligned}
\mu_{K_1} &= 7195.29 \text{ N/mm}, & \sigma_{K_1} &= 364.86 \text{ N/mm}, \\
\mu_{K_2} &= 4116.43 \text{ N/mm}, & \sigma_{K_2} &= 205.63 \text{ N/mm}, \\
\rho_K &= 0.3051, \\
\mathbf{\Sigma}_K &= \begin{pmatrix} 133121 & 22894 \\ 22894 & 42283 \end{pmatrix} (\text{N/mm})^2.
\end{aligned} \tag{21}$$

Kolmogorov–Smirnov tests confirm excellent agreement with the normal assumption (p-values > 0.77 for both margins). The positive but moderate correlation ($\rho_K = 0.305$) reflects co-variation between storey stiffnesses from the same structural system.

3.3.2 Joint Lognormal Fit for Loads (P_1, P_2)

The lognormal parameters are derived from the sample moments via

$$\zeta_i = \sqrt{\ln(1 + \text{COV}_{P_i}^2)}, \quad \lambda_i = \ln(\mu_{P_i}) - \frac{1}{2}\zeta_i^2, \tag{22}$$

with the Nataf approximation for the log-space correlation. The fitted parameters are:

$$\begin{aligned}
\mu_{P_1} &= 100,089 \text{ N}, & \sigma_{P_1} &= 29,974 \text{ N}, & \text{COV}_{P_1} &= 0.300, \\
\mu_{P_2} &= 301,457 \text{ N}, & \sigma_{P_2} &= 90,208 \text{ N}, & \text{COV}_{P_2} &= 0.299, \\
\rho_P &= 0.485, \\
\lambda_{P_1} &= 11.471, & \zeta_{P_1} &= 0.2931, & \lambda_{P_2} &= 12.574, & \zeta_{P_2} &= 0.2928, \\
\rho_{\ln P} &= 0.4955 \text{ (Nataf)}.
\end{aligned} \tag{23}$$

KS tests confirm lognormal adequacy (p-values > 0.88 for both variables).

Figure 1 — Stiffness Data (K_1, K_2) :
Marginal Histograms and Joint Scatter with Bivariate Normal Contours

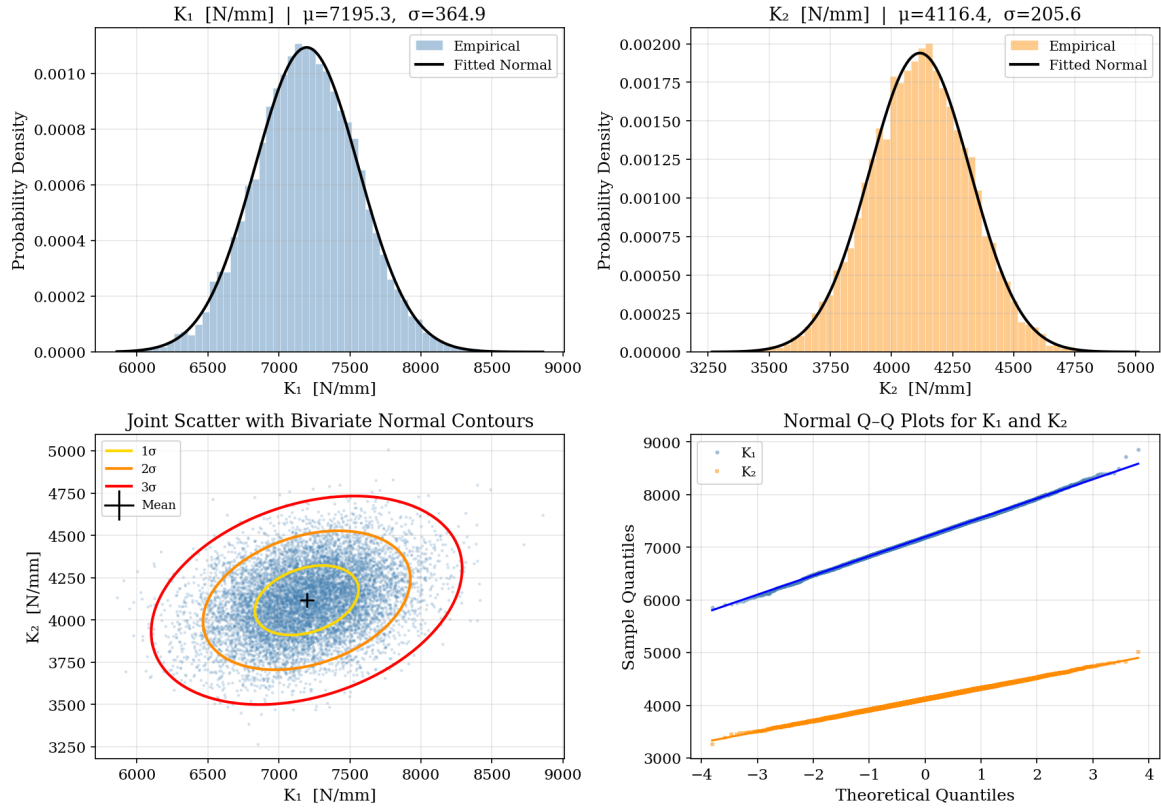


Figure 5. Stiffness data (K_1, K_2): marginal histograms with fitted normal densities (top), joint scatter with bivariate normal confidence ellipses at $1\sigma, 2\sigma, 3\sigma$ (bottom left), and normal Q-Q plots for both margins (bottom right). All panels confirm the adequacy of the bivariate normal model.

Figure 2 — Load Data (P_1, P_2) :
Marginal Histograms and Joint Scatter with Joint Lognormal Contours

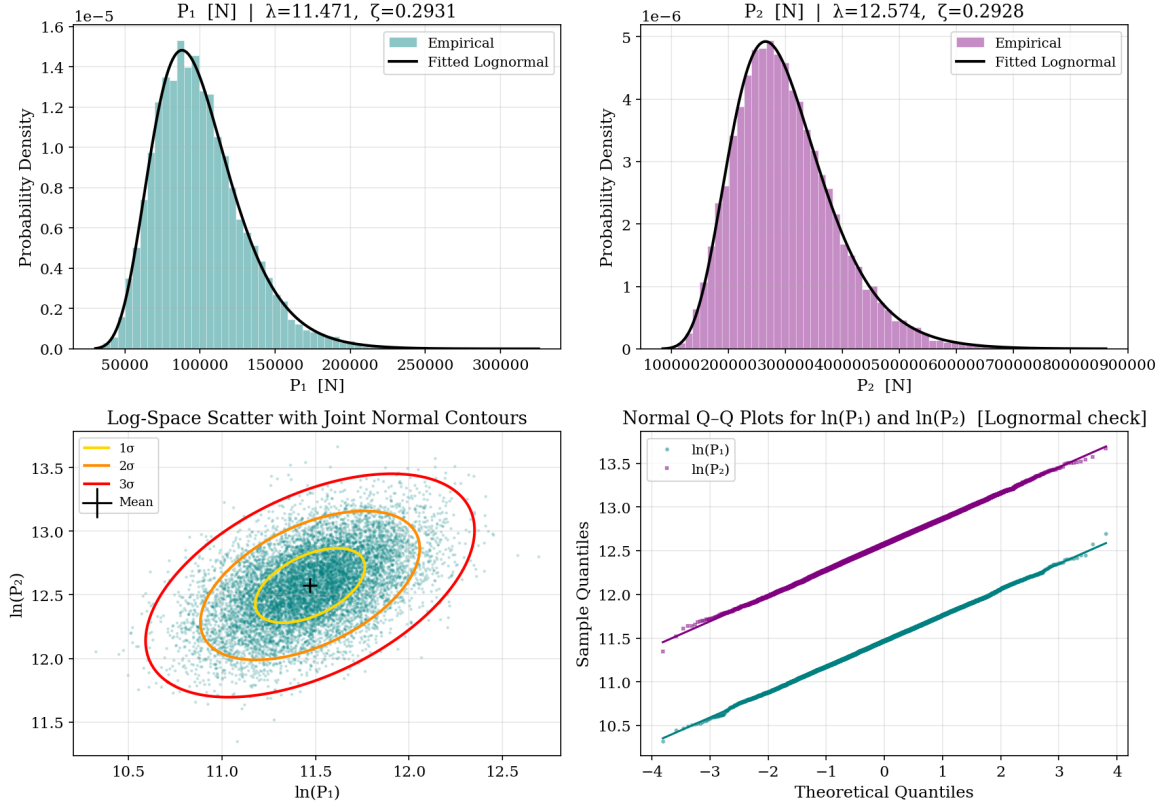


Figure 6. Load data (P_1, P_2): marginal histograms with fitted lognormal densities (top), log-space scatter with joint normal confidence ellipses (bottom left), and Q-Q plots in log-space confirming lognormality (bottom right).

3.4 Case 1: HLRF with Fixed Nominal Loads

When the loads are fixed at their nominal values $P_1^{\text{nom}} = 2.04 \times 10^5$ N and $P_2^{\text{nom}} = 6.12 \times 10^5$ N, the reliability problem involves only the two stiffness variables (K_1, K_2).

3.4.1 Transformation to Standard Normal Space

Since (K_1, K_2) are jointly normal, the transformation uses the Cholesky factor $\mathbf{L}_K$ of the covariance matrix (21):

$$\mathbf{u}_K = \mathbf{L}_K^{-1} (K - \boldsymbol{\mu}_K), \quad K = \boldsymbol{\mu}_K + \mathbf{L}_K \mathbf{u}_K. \quad (24)$$

The gradient of the limit-state function in physical space is

$$\nabla_{\mathbf{x}} g = \left(\frac{P_1^{\text{nom}} + P_2^{\text{nom}}}{3K_1^2}, \frac{P_2^{\text{nom}}}{3K_2^2} \right)^T, \quad (25)$$

and in standard normal space $\nabla_{\mathbf{u}} G = \mathbf{L}_K^T \nabla_{\mathbf{x}} g$.

3.4.2 HLRF Iteration History and Results

Table 6. HLRF iteration history, Case 1 (fixed loads, bivariate normal (K_1, K_2)).

Iter.	K_1 [N/mm]	K_2 [N/mm]	u_1	u_2	$g(K)$	$ \nabla G $	β
0	7195.29	4116.43	0.00000	0.00000	12.6400	3.5635	-3.5470
1	6224.80	3490.02	-2.65991	-2.34656	-2.1485	4.8789	-3.1063
2	6356.21	3563.23	-2.29976	-2.08813	-0.0443	4.6800	-3.0969
3	6358.83	3564.88	-2.29257	-2.08198	-0.0000	4.6759	-3.0969
4	6358.81	3564.90	-2.29264	-2.08190	≈ 0	4.6759	-3.0969
7	6358.80	3564.90	-2.29265	-2.08188	≈ 0	4.6759	3.0969

The converged results are:

$$\begin{aligned}
 K_1^* &= 6358.80 \text{ N/mm}, & K_2^* &= 3564.90 \text{ N/mm}, \\
 \mathbf{u}^* &= (-2.2927, -2.0819), & \beta &= 3.097, & P_f^{\text{FORM}} &= 9.78 \times 10^{-4}.
 \end{aligned} \quad (26)$$

Both stiffness components at the design point are below their respective means, reflecting the physically intuitive result that failure occurs when the structure is less stiff than expected. The sensitivity direction $\boldsymbol{\alpha} = (-0.740, -0.672)$ indicates that K_1 and K_2 contribute nearly equally to the failure event, with K_1 slightly dominant.

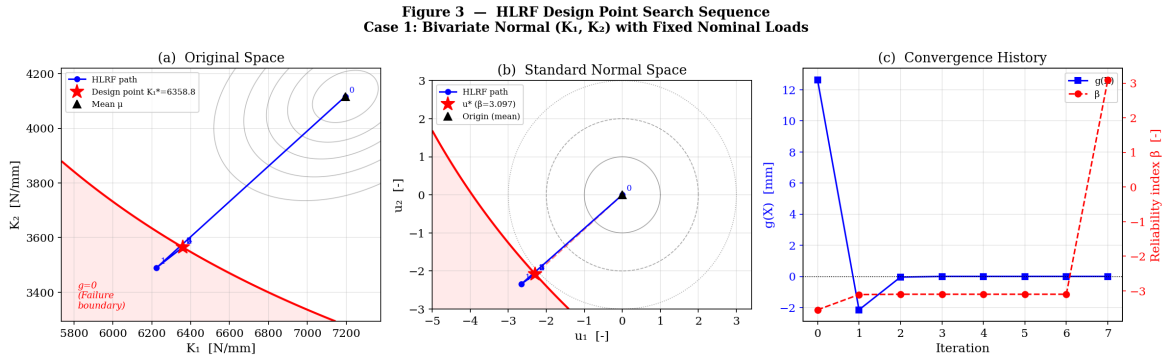


Figure 7. HLRF design point search sequence for Case 1 (fixed loads). *Left:* physical K_1 – K_2 space with the limit-state boundary (red), failure domain (shaded), and HLRF path. *Centre:* standard normal space (u_1, u_2) with iso-probability circles and the design point $\mathbf{u}^*$ on the $\beta = 3.097$ circle. *Right:* convergence of $g(K)$ and β with iteration.

3.5 Case 2: HLRF with All Four Random Variables

When the loads P_1 and P_2 are treated as random (jointly lognormal), the state vector is $\mathbf{X} = (K_1, K_2, P_1, P_2)^\top$ and the standard normal space is four-dimensional: $\mathbf{u} = (u_{K_1}, u_{K_2}, u_{P_1}, u_{P_2})^\top$.

3.5.1 Block Transformation

Since K and P are assumed mutually independent, the transformation is block-diagonal:

$$\mathbf{u}_K = \mathbf{L}_K^{-1}(K - \boldsymbol{\mu}_K), \quad \mathbf{u}_P = \mathbf{L}_{\ln P}^{-1}(\ln P - \boldsymbol{\mu}_{\ln P}). \quad (27)$$

The gradient is computed by central finite differences and the HLRF update rule (3) is applied in $\mathbb{R}^4$.

3.5.2 HLRF Iteration History and Results

Table 7. HLRF iteration history, Case 2 (all four random variables). Selected iterations.

Iter	K_1	K_2	P_1 [N]	P_2 [N]	u_1	u_2	u_3	u_4	g	β
0	7195.3	4116.4	100,089	301,457	0.000	0.000	0.147	0.085	56.99	-4.86
1	7002.4	3991.9	216,848	1,173,697	-0.529	-0.467	2.785	3.924	-64.20	-3.50
2	7060.3	4023.4	166,370	793,978	-0.370	-0.357	1.880	2.903	-11.12	-3.15
3	7073.6	4033.4	158,327	717,344	-0.334	-0.317	1.711	2.600	-0.548	-3.13
4	7074.0	4034.2	158,240	713,271	-0.332	-0.314	1.709	2.579	-0.002	-3.13
9	7074.0	4034.2	158,281	713,245	-0.332	-0.313	1.710	2.578	≈ 0	3.128

The converged results are:

$$\begin{aligned} K_1^* &= 7074.0 \text{ N/mm}, & K_2^* &= 4034.2 \text{ N/mm}, \\ P_1^* &= 158,281 \text{ N}, & P_2^* &= 713,245 \text{ N}, \\ \mathbf{u}^* &= (-0.332, -0.313, 1.710, 2.578), \\ \beta &= 3.128, & P_f^{\text{FORM}} &= 8.81 \times 10^{-4}. \end{aligned} \quad (28)$$

A key observation is that the stiffness design point values ($K_1^* = 7074$, $K_2^* = 4034$) are very close to their means, indicating that the failure event in Case 2 is primarily driven by the loads. The load design point values ($P_1^* = 158,281$ N and $P_2^* = 713,245$ N) are 55% and 137% above the nominal values respectively, confirming that the system's response to load variability dominates the reliability. The larger standard normal components $u_3 = 1.710$ and $u_4 = 2.578$ (for the loads) relative to $u_1 = -0.332$ and $u_2 = -0.313$ (for the stiffnesses) reinforce this interpretation.

Comparing Case 1 and Case 2:

- The reliability index increases slightly from $\beta = 3.097$ to $\beta = 3.128$ (failure probability decreases from 9.78×10^{-4} to 8.81×10^{-4}) when load variability is included.

- This counter-intuitive result occurs because the nominal loads used in Case 1 ($P_1^{\text{nom}} = 2.04 \times 10^5$, $P_2^{\text{nom}} = 6.12 \times 10^5$) are already in the upper range relative to the fitted lognormal means ($\mu_{P_1} = 1.00 \times 10^5$, $\mu_{P_2} = 3.01 \times 10^5$): the nominal loads are higher than the actual mean loads, making Case 1 slightly more conservative.

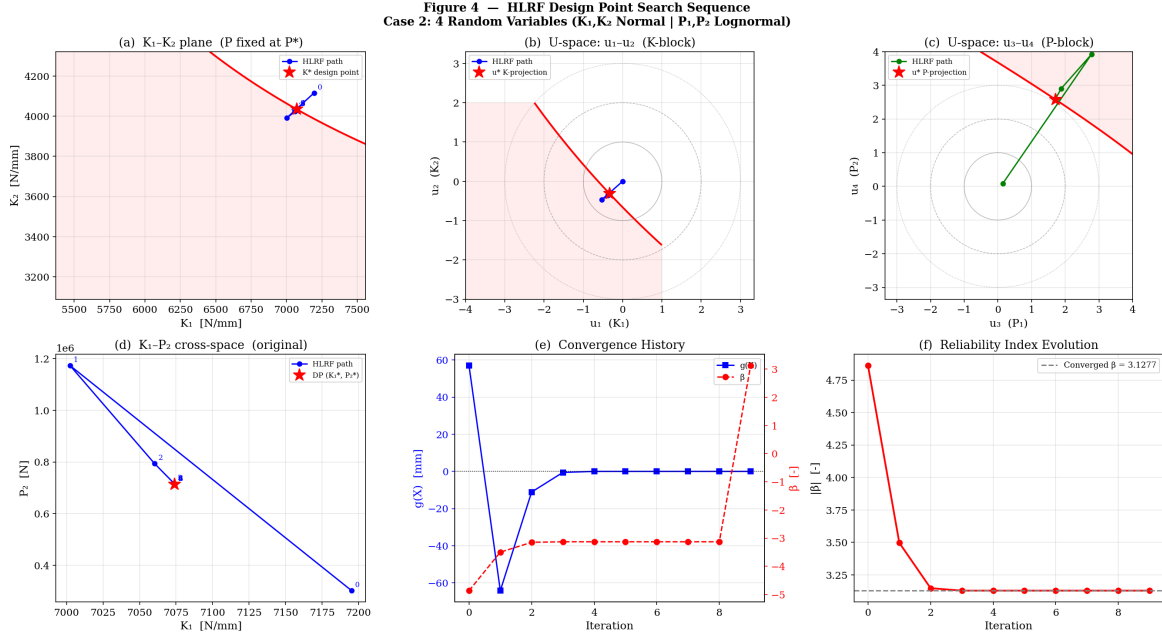


Figure 8. HLRF design point search sequence for Case 2 (four random variables). Subplots show projections of the 4D search path: (a) K_1 – K_2 plane (original space), (b) u_1 – u_2 standard normal space (stiffness block), (c) u_3 – u_4 standard normal space (load block), (d) K_1 – P_2 cross-plane, (e) convergence of $g(X)$, and (f) evolution of the reliability index β .

3.6 Monte Carlo Simulation

Crude Monte Carlo simulation ($N = 500,000$), importance sampling centred at the design point ($N = 100,000$), and importance sampling with optimised covariance ($N = 100,000$ after a pilot covariance update) are applied to both Problem 2 cases. The adaptive importance sampling method is not included for Problem 2, following the scope restriction for this part of the analysis.

Results are summarised in Table 8. The reliability-index estimates are compared in Figure 9, while the corresponding probability of failure estimates are compared in Figure 10.

Table 8. Monte Carlo and importance sampling results for Problem 2. FORM references: Case 1: $\beta = 3.097$, $P_f = 9.78 \times 10^{-4}$; Case 2: $\beta = 3.128$, $P_f = 8.81 \times 10^{-4}$.

Case	Method	N	$\hat{P}_f$	$\hat{\beta}$	CoV	95% CI
Case 1	FORM (reference)	—	9.78×10^{-4}	3.097	—	—
	Crude MCS	500,000	1.148×10^{-3}	3.049	0.042	$[1.054, 1.242] \times 10^{-3}$
	IS at design point	100,000	1.063×10^{-3}	3.072	0.006	$[1.051, 1.076] \times 10^{-3}$
	IS with optimised covariance	100,000	1.063×10^{-3}	3.072	0.006	$[1.051, 1.076] \times 10^{-3}$
Case 2	FORM (reference)	—	8.81×10^{-4}	3.128	—	—
	Crude MCS	500,000	8.840×10^{-4}	3.127	0.048	$[8.016, 9.664] \times 10^{-4}$
	IS at design point	100,000	9.162×10^{-4}	3.116	0.006	$[9.055, 9.269] \times 10^{-4}$
	IS with optimised covariance	100,000	9.250×10^{-4}	3.113	0.006	$[9.13, 9.37] \times 10^{-4}$

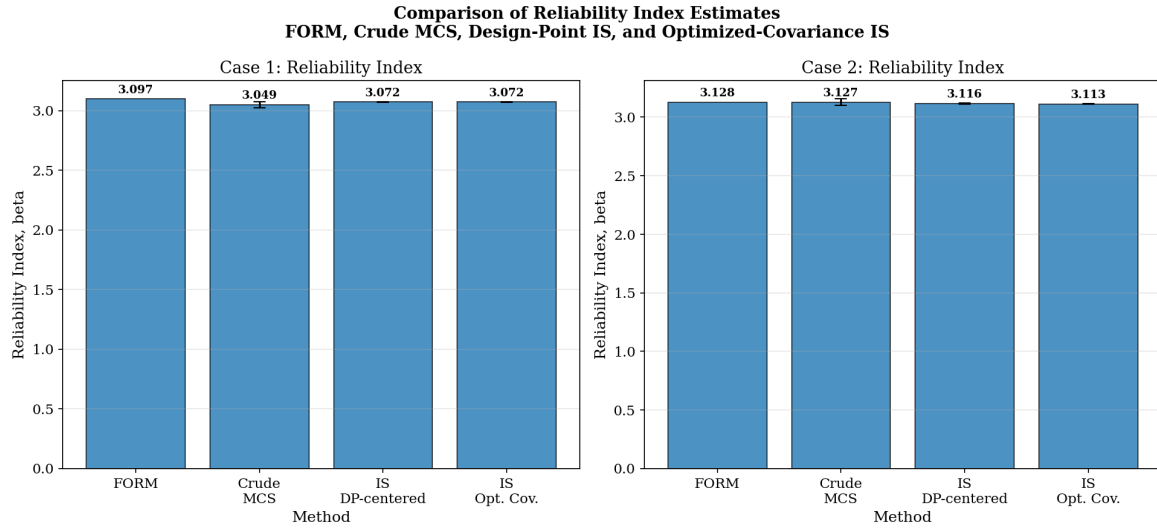


Figure 9. Comparison of reliability-index estimates $\hat{\beta}$ for Problem 2. Error bars represent the transformed 95% confidence intervals from the probability of failure estimates. Both importance sampling methods give estimates close to the FORM reference values, while crude Monte Carlo has larger uncertainty.

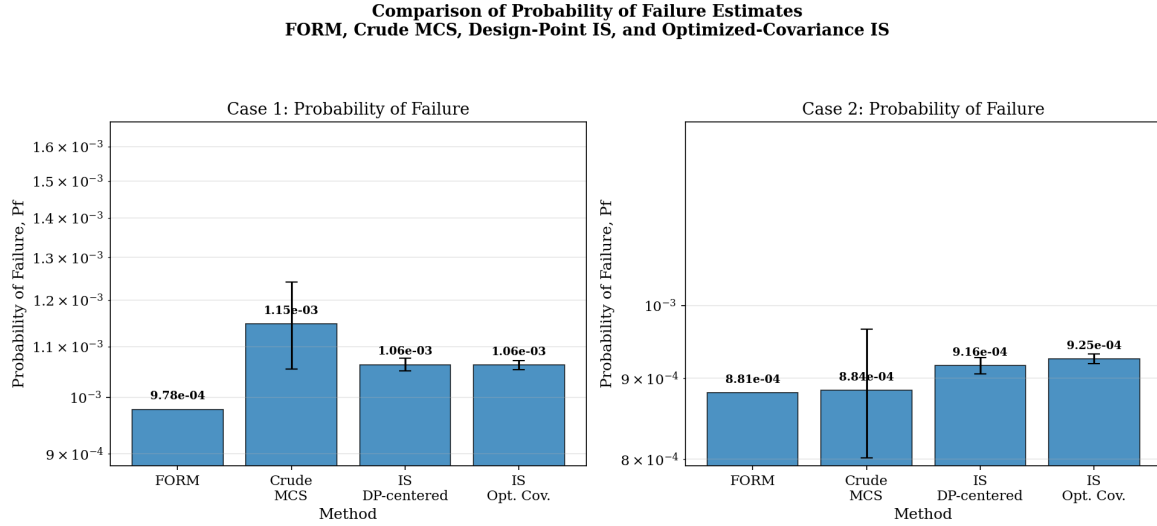


Figure 10. Comparison of probability of failure estimates $\hat{P}_f$ for Problem 2. Error bars represent 95% confidence intervals. The importance sampling methods substantially reduce the coefficient of variation relative to crude Monte Carlo, from approximately 4–5% to below 1%.

3.6.1 Discussion of Simulation Results

The simulation results are consistent with the FORM reference values for both Problem 2 cases. A few observations are noteworthy:

1. **Crude Monte Carlo** gives estimates close to the FORM values for both cases. Since the failure probabilities are of order 10^{-3} , the crude Monte Carlo simulation collects several hundred failure samples in 500,000 draws. This gives coefficients of variation of approximately 4–5%, which is acceptable but still less efficient than importance sampling.
2. **Importance sampling at the design point** gives much narrower confidence intervals using only 100,000 samples. This is because the proposal distribution is shifted toward the design point, concentrating samples near the most probable failure region instead of wasting most samples in the safe domain.
3. **Importance sampling with optimised covariance** gives similar estimates to the design-point-centred IS method, while also adapting the proposal covariance in standard normal space. For Problem 2, the improvement over basic IS is modest because the failure boundary is already nearly linear near the design point and the design-point proposal is already effective.
4. The agreement between FORM and the simulation methods confirms that the linearisation of the limit-state function at the design point is a good approximation for both Problem 2 cases. In Case 1, failure is mainly caused by unusually low stiffness values. In Case 2, the design point shows that failure is driven mainly by high load realisations, especially P_2 .

4 Comparative Discussion and Conclusions

4.1 Sensitivity to Probabilistic Modelling Assumptions

The two problems together demonstrate that structural reliability estimates are highly sensitive to the assumed probability model. For Problem 1, the ratio of failure probabilities between the Gaussian and non-Gaussian copula models exceeds 500, even though all first-two moments and the correlation coefficient are identical. For Problem 2, the ratio between Cases 1 and 2 is modest (approximately 1.1), but this reflects the particular choice of nominal loads relative to the fitted mean loads.

4.2 FORM Accuracy

FORM performs well in all cases examined. For Problem 1 Model A, the IS estimates are slightly below the FORM value ($\hat{\beta} \approx 4.06$ vs $\beta = 4.027$), indicating mild curvature of the failure boundary that FORM does not capture. For Problem 1 Model B and both cases of Problem 2, the agreement between FORM and IS is excellent, suggesting that the failure boundaries are nearly planar in the neighbourhood of the design points.

4.3 Efficiency of Simulation Methods

Table 9 summarises the qualitative performance of each simulation strategy across the two problems.

Table 9. Qualitative summary of simulation method performance.

Method	Strengths	Limitations
Crude Monte Carlo	Unbiased; simple to implement; does not require a design point.	Inefficient for rare events; requires many samples to obtain small confidence intervals.
IS at design point	Low CoV; concentrates samples near the most probable failure region; efficient for FORM-like problems.	Requires an accurate design point before simulation.
IS with optimised covariance	Further improves the proposal distribution by adapting covariance in standard normal space.	Requires a pilot run and may offer only modest improvement when the basic design-point IS is already effective.

4.4 Conclusions

This study has demonstrated a complete reliability analysis workflow for two structural problems using both analytical (FORM/HLRF) and simulation-based methods.

For **Problem 1**, the HLRF algorithm converges reliably for both probabilistic models. The jointly Gaussian model gives $\beta = 4.027$ and $P_f = 2.83 \times 10^{-5}$, while the lognormal–Gumbel copula model gives $\beta = 5.313$ and $P_f = 5.39 \times 10^{-8}$, a difference of over two orders of magnitude. IS centred at the design point provides the most consistent simulation estimates, particularly for the non-Gaussian model where crude CMC yields no failures in 500,000 samples.

For **Problem 2**, the fitted bivariate normal distribution (KS p-values > 0.77) and joint lognormal distribution (KS p-values > 0.88) adequately capture the monitoring data. HLRF converges in 7 iterations for the 2D problem and 9 iterations for the 4D problem. The resulting FORM estimates ($\beta = 3.097$ and $\beta = 3.128$) are confirmed by Monte Carlo simulation and importance sampling. The analysis reveals that load variability, when properly characterised through the fitted lognormal model, has a modest effect on the estimated failure probability relative to the fixed-load case, because the nominal loads specified in the problem statement are already substantially higher than the fitted mean loads.

Overall, the results confirm that both the choice of probabilistic model and the selection of a variance-reduction strategy are critical considerations in practical structural risk assessment, particularly when dealing with rare failure events.

A Consolidated FORM Results

Table A1. Consolidated FORM results for all analyses.

Problem	Case	β	P_f^{FORM}	$\mathbf{u}^*$	$\mathbf{x}^*$
1	Model A (Gaussian)	4.027	2.83×10^{-5}	(−3.628, 1.748)	(5.744, 16.498)
1	Model B (Lognormal–Gumbel)	5.313	5.39×10^{-8}	(−2.567, 4.652)	(8.677, 37.646)
2	Case 1 (fixed loads, 2D)	3.097	9.78×10^{-4}	(−2.293, −2.082)	see Eq. (26)
2	Case 2 (random loads, 4D)	3.128	8.81×10^{-4}	(−0.332, −0.313, 1.710, 2.578)	see Eq. (28)

B Problem 2: Distribution Parameter Summary

Table A2. Fitted distribution parameters for Problem 2 (from 10,000-sample subset).

Variable	Distribution	μ	σ	Additional params	KS p-value
K_1 [N/mm]	Normal	7195.29	364.86	$\rho_K = 0.305$	0.776
K_2 [N/mm]	Normal	4116.43	205.63	—	0.922
P_1 [N]	Lognormal	100,089	29,974	$\lambda = 11.471, \zeta = 0.293$	0.997
P_2 [N]	Lognormal	301,457	90,208	$\lambda = 12.574, \zeta = 0.293$	0.884